

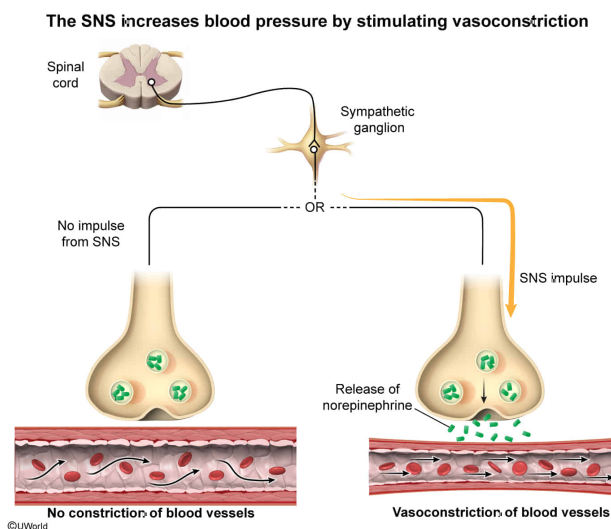
Presumably, activation of the sympathetic nervous system increases blood pressure during times of stress because:

- ✔ ☒ A. norepinephrine, acting as a neurotransmitter, promotes vasoconstriction of blood vessels. [78%]
- ☐ B. norepinephrine, acting as a neurotransmitter, promotes vasodilation of blood vessels. [17%]
- ☐ C. norepinephrine suppresses the renin-angiotensin-aldosterone system in the kidneys. [3%]
- ☐ D. norepinephrine inhibits the release of antidiuretic hormone. [1%]

Correct

77% answered correctly

Explanation:



The **autonomic nervous system (ANS)**, a subset of the **peripheral nervous system**, regulates the activities of the internal organs and endocrine system at rest and in times of stress. The ANS is further divided into the **sympathetic nervous system (SNS)** and **parasympathetic nervous system (PNS)**.

The SNS, which dominates in times of stress, coordinates the body's response to a perceived threat. The SNS mobilizes resources toward energy expenditure and suspends activities such as digestion and excretion whereas the PNS acts in opposition to the SNS when no threat is perceived. These two divisions innervate many of the same targets, often with antagonistic effects.

The synapses of the SNS can be classified as either **cholinergic** (releasing **acetylcholine (ACh)**), or **adrenergic** (releasing **norepinephrine (NE)**). All **preganglionic neurons** of the ANS release ACh. **Postganglionic neurons** of the

PNS also release ACh, but postganglionic neurons of the SNS generally release NE instead.

The question asks how the SNS can increase blood pressure during stress.

Sympathetic nerves release the neurotransmitter NE, which acts directly on the smooth muscle of **blood vessels** and **causes them to constrict**. This **vasoconstriction** increases blood pressure by decreasing the volume inside blood vessels. When this volume is lowered, the force of blood flow is raised, thereby increasing blood pressure.

(Choice B) NE promotes vasoconstriction (*not* vasodilation) of blood vessels. Vasodilation would decrease (*not* increase) blood pressure, an opposite effect of SNS stimulation.

(Choice C) The SNS can activate the **renin-angiotensin-aldosterone system** (RAAS) in the kidneys, which ultimately results in vasoconstriction and an increase in blood pressure. Suppressing the RAAS would result in a decrease (*not* increase) in blood pressure.

(Choice D) **Antidiuretic hormone** (ADH) increases blood pressure by increasing the reabsorption of water in the kidneys, as well as promoting vasoconstriction of blood vessels. Inhibiting ADH release would decrease (*not* increase) blood pressure.

Educational objective:

In times of stress, the sympathetic nervous system (SNS) mobilizes the body's resources to deliver blood toward vital organs and away from extremities. One way that SNS activation can increase blood pressure is by releasing the neurotransmitter norepinephrine to directly stimulate the constriction of vascular smooth muscle. The SNS can also increase blood pressure by activating the renin-angiotensin-aldosterone system in the kidneys.

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Biology

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System:
Endocrine and Nervous Systems